

AC9M6A01 • Year 6 Mathematics

# Growing Patterns and Rational-number Rules

Connect visual growth, term values and direct or recursive rules

## We are learning to...

Students identify what changes and what remains constant, represent patterns in tables and diagrams, use recursive and direct rules and test whether a rule generates every term.

Represent

Reason

Calculate

Interpret

Verify

## Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

## Curriculum focus

recognise and use rules that generate visually growing patterns and number patterns involving rational numbers

# Connect a growing tile pattern to a rule

| term n | 1 | 2  | 3  | 4  | n           |
|--------|---|----|----|----|-------------|
| tiles  | 5 | 8  | 11 | 14 | $3n + 2$    |
| growth |   | +3 | +3 | +3 | constant +3 |

The recursive rule adds 3 each term. The direct rule  $3n + 2$  gives any term without generating all previous terms.

## Extend patterns involving fractions and decimals

**0.5, 1.0, 1.5, 2.0**

add 0.5; term =  $0.5n$

**$\frac{3}{4}$ , 1,  $1\frac{1}{4}$ ,  $1\frac{1}{2}$**

add  $\frac{1}{4}$ ; term =  $\frac{n}{4} + \frac{1}{2}$

**visual border growth**

count repeated section plus  
fixed corners

**digital table**

test many terms after defining  
rule

A rule must fit the visual structure and all known terms. Equivalent forms of a rational rule may look different but produce the same values.

### Build and annotate the model

Represent growing patterns and rational-number rules and label the important parts, quantities or choices.

### Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

### Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

## Common mix-ups

- **Rule describes only the next term:** A direct rule should work for any term number.
- **Constant part omitted:** Visual patterns often contain repeated growth plus fixed pieces.
- **Rule accepted from two terms only:** Test several terms and the visual structure.
- **Term number confused with term value:** Keep input  $n$  separate from output.

## Quick check

- Find the next two terms.
- Write a recursive rule.
- Test  $3n + 2$ .
- Describe fixed and growing parts.
- Extend a decimal pattern.

**Teacher decision:** continue when students explain the model and verify a new example; otherwise return to Slide 2.